📊 HR Analytics Dashboard Report

Project Title

HR Analytics Dashboard Using Power BI

Objective

The objective of this project is to analyze organizational HR data to derive meaningful insights related to employee attrition, workforce distribution, and job satisfaction.

This dashboard aims to support **data-driven HR decision-making** by identifying patterns in employee behavior, retention, and demographic structure.

Problem Statement

Organizations often face challenges in understanding:

- Why employees leave the organization
- Which departments have higher attrition rates
- How demographics influence employee retention
- How job satisfaction varies across roles

This project addresses these challenges by visualizing HR data in an interactive dashboard.

Dataset Description

The dataset used in this project is an **embedded HR dataset** within Power BI.

Key Fields:

- Employee Count
- Attrition Count
- Attrition Rate
- Active Employees
- Age
- Age Band
- Gender

- Department
- Job Role
- Education Field
- Job Satisfaction

Note: No external database connection is required.

Tools & Technologies Used

- Power BI Desktop
- Power Query (Data Transformation)
- DAX (Data Analysis Expressions)
- Data Visualization Techniques

Data Preparation & Transformation

The following preprocessing steps were performed:

- Removal of null or inconsistent values
- Creation of calculated fields (e.g., Attrition Rate)
- Age grouping into Age Bands
- Data type standardization
- Categorization of Education Field and Job Roles

These steps ensured clean and structured data for visualization.

Data Modeling Approach

A **flat data model structure** was used since the dataset is embedded.

Key relationships were maintained implicitly within Power BI visuals.

Dashboard Overview

The dashboard is designed as a **single-page interactive report** consisting of:

📌 **KPI Cards**

- Total Employee Count

- Attrition Count
- Attrition Rate
- Active Employees
- Average Age

📊 **Visualizations**

- **Pie Chart** → Attrition by Department
- **Bar Chart** → Attrition by Education Field
- **Column Chart** → Age Band and Gender distribution
- **Donut Charts** → Attrition by Gender and Age Group
- **Matrix Table** → Job Satisfaction by Job Role

🎛️ **Filters**

- Education Field slicer for dynamic filtering

Key Insights

The analysis of the HR dataset reveals the following insights:

1. Attrition Patterns

Certain departments show higher attrition rates compared to others, indicating potential organizational or managerial issues.

2. Demographic Influence

Younger age groups tend to exhibit higher attrition rates, suggesting early-career mobility.

3. Education Impact

Employees from specific education backgrounds show varying levels of retention and job stability.

4. Job Satisfaction Trends

Job satisfaction varies significantly across job roles, which may influence employee retention.

5. Workforce Composition

The organization maintains a balanced distribution of employees across gender and age groups, with minor variations.

Business Impact

This dashboard enables HR teams to:

- Identify risk areas for employee turnover
- Improve retention strategies
- Optimize workforce planning
- Monitor employee satisfaction trends
- Support strategic HR decisions with data

Conclusion

The HR Analytics Dashboard successfully transforms raw HR data into actionable insights.

It provides a clear understanding of workforce behavior, attrition trends, and employee satisfaction levels.

Such dashboards are essential for modern HR analytics, enabling organizations to make informed and data-driven decisions.

Future Enhancements

The project can be extended with:

- Predictive analytics for attrition forecasting (Machine Learning)
- Time-series analysis of employee trends
- Drill-through pages for department-level deep analysis
- Integration with live HR databases
- Role-based access dashboards for management